Part IA: Mathematics for Natural Sciences B
Examples Sheet 14: Contour sketching, and constrained optimisation

Model Solutions

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Classification of stationary points

1. Describe the classification of stationary points of a function of two variables $f(x, y)$ via conditions on the second partial derivatives of the function at the stationary point. Hence or otherwise, sketch the contours of the functions:

(a) $f(x, y) = x^2 - 4xy + y^2$;

(b) $f(x, y) = x^2 + xy + y^2$;

(c) $f(x, y) = x^2 + 2xy + y^2$.

Add some arrows to your sketches, indicating the direction of the gradient vector at salient points.

Solution: A *stationary point* of a multivariable function $f(x, y)$ is a point where $\nabla f = 0$, that is, the gradient vanishes. Such points can be *maxima*, *minima* or *saddles*, but importantly: these are *not* the only possibilities! Writing f_{xx}, f_{xy}, f_{yy} for the appropriate second derivatives of f evaluated at the stationary points, we can tell if a point is a maximum, minimum, or saddle, using the *second derivative test*:

- If $f_{xx}f_{yy} - f_{xy}^2 < 0$, the point is a saddle.
- If $f_{xx}f_{yy} - f_{xy}^2 > 0$ and $f_{xx} + f_{yy} > 0$, the point is a minimum.
- If $f_{xx}f_{yy} - f_{xy}^2 > 0$ and $f_{xx} + f_{yy} < 0$, the point is a maximum.
- Otherwise, the test is *inconclusive* (the point can still be a minimum, maximum or saddle, or something more interesting).

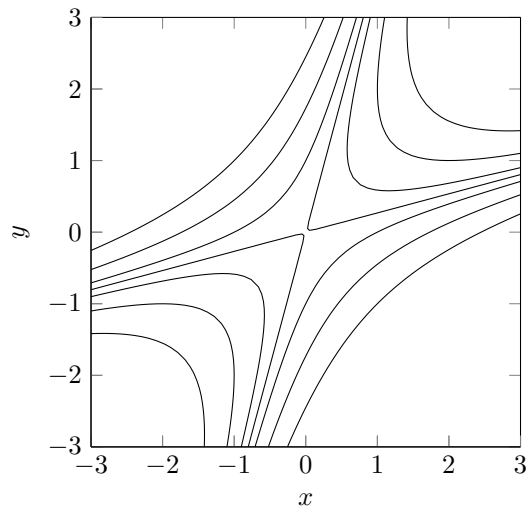
Applying this test to the given functions, we have:

(a) $\nabla f = (2x - 4y, -4x + 2y) = (0, 0)$ at stationary points. Solving simultaneously, we see that $(x, y) = (0, 0)$ is the only solution. Taking second derivatives, we have $f_{xx} = 2$, $f_{xy} = -4$ and $f_{yy} = 2$. Since $f_{xx}f_{yy} - f_{xy}^2 = -16 < 0$, the stationary point is a saddle.

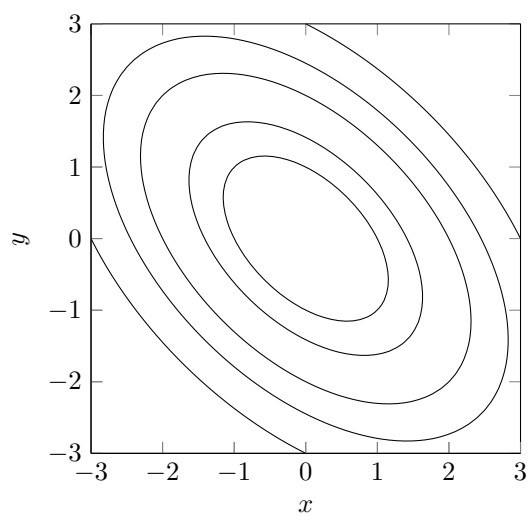
We can be more precise about the contour lines that pass through the saddle point. At the point $(0, 0)$, we have that the contour lines will satisfy:

$$x^2 - 4xy + y^2 = 0.$$

Rearranging for y, we see that $y = (2 \pm \sqrt{3})x$, giving an exact equation for the contour lines passing through $(0, 0)$.



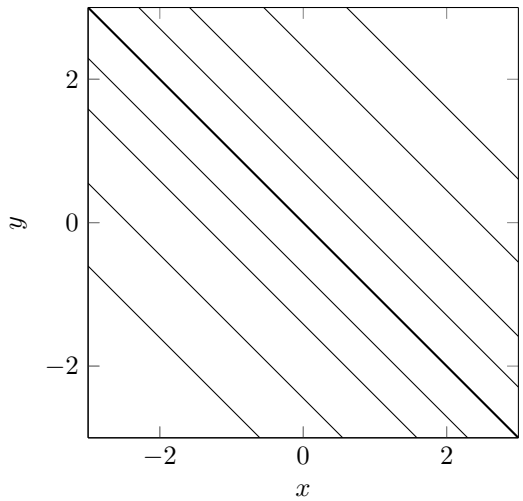
(b) $\nabla f = (2x + y, x + 2y) = (0, 0)$ at stationary points. Solving simultaneously, we see that $(x, y) = (0, 0)$ is the only solution. Taking second derivatives, we have $f_{xx} = 2$, $f_{xy} = 1$, $f_{yy} = 2$. Since $f_{xx}f_{yy} - f_{xy}^2 = 3 > 0$ and $f_{xx} + f_{yy} > 0$, this point is a minimum.



(c) $\nabla f = (2x + 2y, 2x + 2y) = (0, 0)$. Solving simultaneously, we see that the general solution is not just a point, but a *line* $y = -x$. Thus we have a line of stationary points (this is not a minimum, maximum or saddle!). We cannot apply the second derivative test here, but we can (luckily) calculate the contours explicitly in this case. Since:

$$x^2 + 2xy + y^2 = c \qquad \Rightarrow \qquad (x + y)^2 = c \qquad \Rightarrow \qquad x + y = \pm\sqrt{c},$$

we see that the general equations of the contour lines are all parallel to the line $y = -x$. We can only have lines where c is non-negative, which means the line of stationary points contains all of the lowest values of the function. The sketch looks like:



Contour sketching

2. (**Large x, y behaviour**) Consider the function:

$$f(x, y) = x^2 + xy + y^2 - x^3.$$

(a) Show that this function has two stationary points, and classify them using the second derivative test. Find also the equations of the contour lines passing through any saddles, if there are any.

(b) Explain why, for very large x, y, the contours become roughly vertical. [*The large x, y behaviour is often a useful point to consider when preparing a contour plot.*]

(c) Hence, sketch the contours of the function.

Solution:

(a) We compute the gradient first, giving $\nabla f = (2x + y - 3x^2, x + 2y) = (0, 0)$ at stationary points. Solving the second equation, we see that $x = -2y$. Substituting into the first equation, we have $y + 4y^2 = 0$, which gives $y = 0$ or $y = -1/4$. Hence the stationary points are:

$$(x, y) = (0, 0), \qquad (1/2, -1/4).$$

The second derivatives are $f_{xx} = 2 - 6x$, $f_{xy} = 1$, $f_{yy} = 2$. For $(0, 0)$, this gives $f_{xx}f_{yy} - f_{xy}^2 = 3 > 0$ and $f_{xx} + f_{xy} = 4 > 0$, so this is a minimum. For $(1/2, -1/4)$, this gives $f_{xx}f_{yy} - f_{xy}^2 < 0$, so this is a saddle.

Taylor expanding near the saddle $(1/2, -1/4)$, we have from our above calculations:

$$\begin{aligned} f(x, y) - f(1/2, -1/4) &= \frac{1}{2}f_{xx}\left(x - \frac{1}{2}\right)^2 + f_{xy}\left(x - \frac{1}{2}\right)\left(y + \frac{1}{4}\right) + \frac{1}{2}f_{yy}\left(y + \frac{1}{4}\right)^2 + \cdots \\ &= -\frac{1}{2}\left(x - \frac{1}{2}\right)^2 + \left(x - \frac{1}{2}\right)\left(y + \frac{1}{4}\right) + \left(y + \frac{1}{4}\right)^2 + \cdots . \end{aligned}$$

Setting $\eta = x - 1/2, \xi = y + 1/4$, we see that the approximate equations of the contour lines passing through the saddle are:

$$-\frac{1}{2}\eta^2 + \eta\xi + \xi^2 = 0.$$

Rearranging, we have:

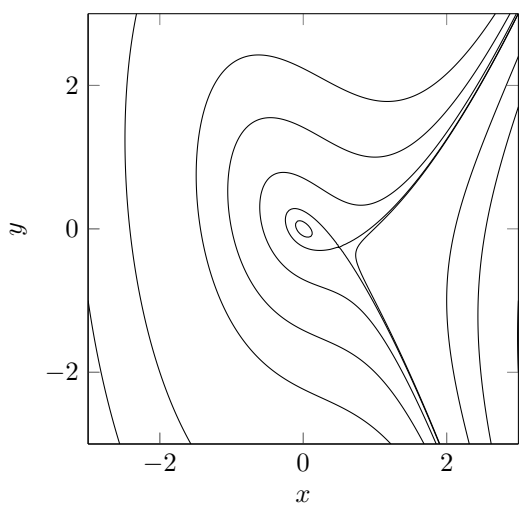
$$\xi = \left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}\right)\eta \qquad \Rightarrow \qquad y + \frac{1}{4} = \left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}\right)\left(x - \frac{1}{2}\right).$$

(b) For very large values of x, y, retaining only the highest order terms, the function becomes:

$$f(x, y) \approx -x^3.$$

The contours of this function are $x =$ const, which are vertical lines.

(c) Putting things together, we see that the contour plot is:



3. (**Symmetry**) Consider the function:

$$f(x, y) = x^2 + 4xy + y^2 + x^4 + y^4.$$

(a) Show that this function has three stationary points, and classify them using the second derivative test. Find also the equations of the contour lines passing through any saddles, if there are any.

(b) Explain why, for very large x, y, the contours begin to resemble $x^4 + y^4 \approx k$ for k constant. What does this figure look like?

(c) Explain why this function is symmetrical in the lines $y = x$ and $y = -x$. [*Symmetry in the lines $x = 0, y = 0$ or the lines $y = x, y = -x$ is often a useful point to consider when preparing a contour plot.*]

(d) Hence, sketch the contours of the function.

Solution:

(a) Taking the gradient, we have:

$$\nabla f = \left(2x + 4y + 4x^3, 4x + 2y + 4y^3\right) = (0, 0),$$

at stationary points. This gives the simultaneous equations:

$$\begin{aligned} 2x + 4y + 4x^3 &= 0, \\ 4x + 2y + 4y^3 &= 0. \end{aligned}$$

Multiplying the first equation by x, and the second by y, we can rearrange them to exploit the symmetry:

$$\begin{aligned} 2x^2 + 4xy + 4x^4 &= 0, \\ 2y^2 + 4xy + 4y^4 &= 0. \end{aligned}$$

Now subtracting the equations, we have $2x^4 - 2y^4 + x^2 - y^2 = 0$. Factorising, we see that $(x^2 - y^2)(2(x^2 + y^2) + 1) = 0$. This gives $x = \pm y$, or $x^2 + y^2 = -\frac{1}{2}$, which is impossible for real x, y.

In the case $x = y$, we have $6x = -4x^3$, which only gives the solution $(x, y) = (0, 0)$. In the case $x = -y$, we have $x = 2x^3$, which gives $x = 0$ or $x = \pm 1/\sqrt{2}$. Hence we have the complete list of stationary points:

$$(0, 0), \qquad \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \qquad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

Taking the second derivative, we have $f_{xx} = 2 + 12x^2$, $f_{xy} = 4$, $f_{yy} = 2 + 12y^2$. This gives:

- $(0, 0)$ has $f_{xx}f_{yy} - f_{xy}^2 = -12 < 0$, so this is a saddle. Near the origin, we can neglect the fourth order powers, which gives the approximate equations of the contour lines as $0 = x^2 + 4xy + y^2$. Rearranging, we see that they are:
$$y = (-2 \pm \sqrt{3})x.$$
- At $(1/\sqrt{2}, -1/\sqrt{2})$, we have $f_{xx}f_{yy} - f_{xy}^2 = 48 > 0$, and $f_{xx} + f_{yy} = 16 > 0$, so this is a minimum.
- At $(-1/\sqrt{2}, 1/\sqrt{2})$, we similarly have a minimum.

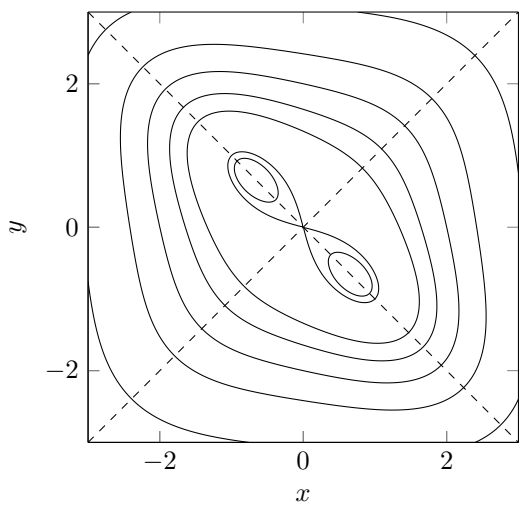
(b) For very large values of x, y, we can neglect the lower order powers of x, y to leave only $f(x, y) \approx x^4 + y^4$. The contours are therefore $x^4 + y^4 \approx k$, for k constant. These are 'circle-like' shapes, but are more 'squared-off' around the corners; for example, we can see the gradient:

$$4x^3 + 4y^3 \frac{dy}{dx} = 0 \qquad \Rightarrow \qquad \frac{dy}{dx} = -\frac{x^3}{y^3},$$

which remains flatter for longer around $x = 0$ than the circle with $dy/dx = -x/y$.

(c) Swapping the variables $x \leftrightarrow y$, we see that the function is unchanged, hence the function must be symmetrical in the line $y = x$. On the other hand, swapping $x \mapsto -y$ and $y \mapsto -x$ gives a reflection in the line $y = -x$, which similarly gives that the function is unchanged, and hence must be symmetrical in the line $y = -x$.

(d) The plot is given by:



4. (**Zero contours**) Consider the function:

$$f(x,y) = \frac{x+y}{x^2+y^2+1}.$$

(a) Show that this function has two stationary points.

(b) Find the equations of the zero contour(s) of the function. Explain also why the function tends to zero for very large x, y. Hence, classify the stationary points you found in (a), and thus sketch the contours of the function. [*If you can calculate them, the zero contours of a function are often a useful tool in classifying stationary points* **without** *using the - rather cumbersome - second derivative test.*]

(c) Now, find the analytical equations of the contours of the function, and hence verify that the sketch you produced in part (b) is accurate. [*If you can calculate them, the analytical equations of the contours can be a very useful check!*]

Solution:

(a) We begin by computing the gradient. We have:

$$\nabla f = \left(\frac{y^2 - 2xy - x^2 + 1}{(x^2+y^2+1)^2}, \frac{x^2 - 2xy - y^2 + 1}{(x^2+y^2+1)^2} \right) = (0,0)$$

at stationary points. These simplify to the simultaneous equations:

$$y^2 - 2xy - x^2 + 1 = 0,$$

$$x^2 - 2xy - y^2 + 1 = 0.$$

Subtracting these equations, we see that $2(y^2 - x^2) = 0$, giving $y = \pm x$. Substituting $y = x$ back into the first equation, we have:

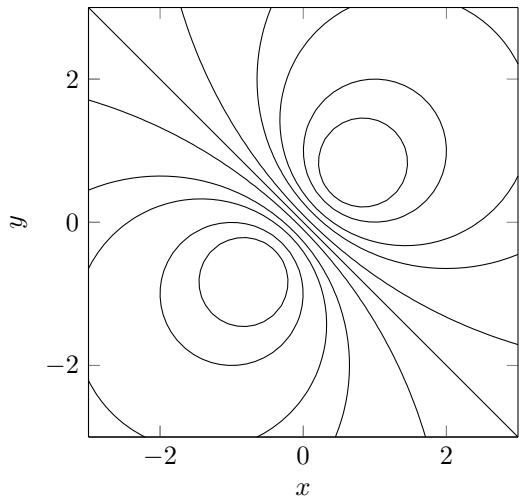
$$1 = 2x^2 \qquad \Rightarrow \qquad x = \pm\frac{1}{\sqrt{2}}.$$

This gives the stationary points $(x,y) = \pm\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. Substituting $y = -x$ back into the first equation, we have $2x^2 + 1 = 0$, which gives no real values for x. Thus these are all the stationary points of the function.

(b) The zero contours of the function are $x + y = 0$, i.e. $y = -x$. Furthermore, for very large values of x, y, the function approaches zero, since the quadratic denominator dominates the linear numerator of the function.

Note that the zero contours are the only places that the function can be zero, and the function is continuous. Since at the point $(x,y) = (1/\sqrt{2}, 1/\sqrt{2})$, the function value is positive, the function must be entirely positive in the region $x + y > 0$. But this entire region is surrounded by places where the function goes to zero - it follows that $(1/\sqrt{2}, 1/\sqrt{2})$ must be a maximum.

Similarly, at $(x,y) = (-1/\sqrt{2}, -1/\sqrt{2})$ the function value is negative, which implies that the function is negative everywhere in the region $x + y < 0$. But then this entire region is surrounded by places where the function goes to zero - it follows that this point must be a minimum. This allows us to produce the contour plot, given overleaf.



(c) The analytical equations of the contours are:

$$\frac{x+y}{x^2+y^2+1} = c,$$

where c is constant. Rearranging, we have:

$$\frac{x+y}{c} = x^2+y^2+1 \qquad \Rightarrow \qquad \left(x-\frac{1}{2c}\right)^2 + \left(y-\frac{1}{2c}\right)^2 = \frac{1}{2c^2} - 1.$$

Hence, the exact contours are in fact circles (with the exception of the case $c = 0$, where $x + y = 0$). All the circles are centred on $(-1/2c, -1/2c)$ as c varies, which is on the line $y = x$.

5. (**Failure of the second derivative test**) Consider the function:

$$f(x, y) = x^2y^2 + x^4 + y^6.$$

(a) Show that this function has a single stationary point, but that the second derivative test fails to classify it.

(b) Using another method, classify it, and sketch the contours of the function.

Solution:

(a) Taking the gradient, we have:

$$\nabla f = (2xy^2 + 4x^3, 2x^2y + 6y^5).$$

Thus at the stationary points, we have:

$$\begin{aligned} 2xy^2 + 4x^3 &= 0, \\ 2x^2y + 6y^5 &= 0. \end{aligned}$$

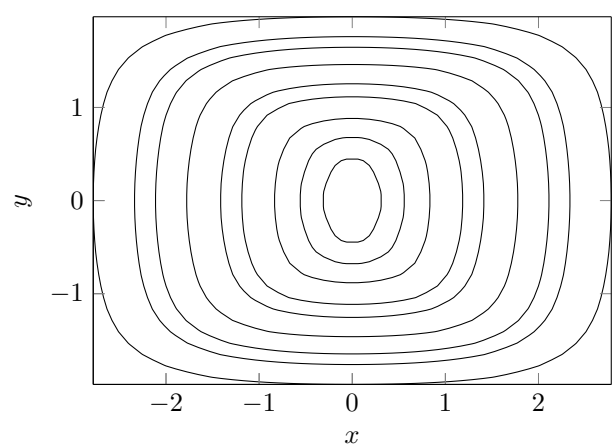
Factoring these equations, we have:

$$\begin{aligned} x(2y^2 + 4x^2) &= 0, \\ y(2x^2 + 6y^4) &= 0. \end{aligned}$$

The first equation implies either $x = 0$ or $(x, y) = (0, 0)$. The second equation implies that either $y = 0$ or $(x, y) = 0$. Thus overall, the only stationary point is $(0, 0)$.

Taking the second derivatives, we have $f_{xx} = 2y^2 + 12x^2$, $f_{yy} = 2x^2 + 30y^4$ and $f_{xy} = 4xy$. These all vanish at $(0, 0)$, which implies the second derivative test is inconclusive.

(b) Note that for all values of $(x, y) \neq (0, 0)$, we have $x^2y^2 + x^4 + y^6 > 0$. Hence the stationary point at $(0, 0)$ must have been a minimum. Far away from $(0, 0)$, the function approaches $f(x, y) \approx y^6$, which gives the contours $y \approx k$, hence the contours become roughly horizontal. Thus we can produce the contour plot:



6. (**Reduction to one dimension, and polar coordinates**) Consider the function:

$$f(x, y) = x^4 + y^4 - x^2y^2.$$

(a) Show that this function has a single stationary point, but that the second derivative test fails to classify it.

(b) By examining the behaviour of the function along the lines $y = x$, $y = -x$, $x = 0$ and $y = 0$, classify the stationary point, and hence sketch the contours of the function. [*Examining a function along specific lines through a stationary point reduces the problem to a one-dimensional problem, often aiding in classification of a point.*]

(c) Show that the function can be rewritten in terms of polar coordinates as:

$$f(r, \theta) = \frac{1}{8}r^4 \left(3\cos(4\theta) + 5\right).$$

Hence, verify that the sketch you provided in part (b) is accurate. [*Sometimes, switching into polar coordinates can be very helpful in understanding the nature of a stationary point - particularly in cases where the function factorises as $g(r)h(\theta)$ in polar form.*]

Solution:

(a) The gradient is:

$$\nabla f = \left(4x^3 - 2xy^2, 4y^3 - 2x^2y\right).$$

Setting this to zero, we have the simultaneous equations:

$$x(4x^2 - 2y^2) = 0,$$
$$y(4y^2 - 2x^2) = 0.$$

The first equation implies that either $x = 0$ or $2x^2 = y^2$, giving $\sqrt{2}x = \pm y$. The second equation implies either $y = 0$ or $\sqrt{2}y = \pm x$.

In the first case $x = 0$, we must have $y = 0$ too. Hence $(0, 0)$ is a stationary point. In the second case, $2x^2 = y^2$, the second equation implies that $4y^2 - y^2 = 0$, giving $y = 0$. Hence $(0, 0)$ is the *only stationary point*.

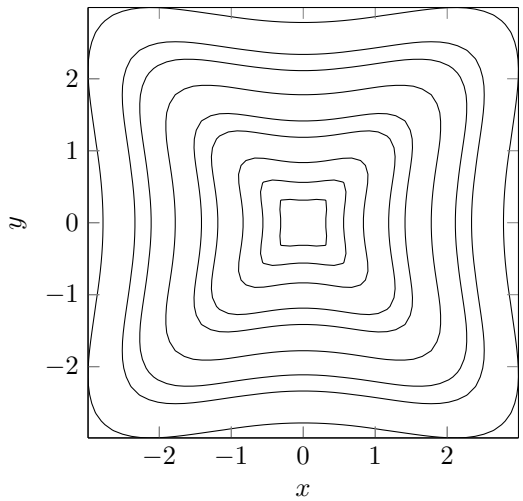
The second derivatives are given by $f_{xx} = 12x^2 - 2y^2$, $f_{xy} = -4xy$ and $f_{yy} = 12y^2 - 2x^2$. These all vanish at $(x, y) = (0, 0)$, and hence the second derivative test is inconclusive.

(b) Doing as the question tells us, we now examine the function along the lines specified. We have:

- $f(x, 0) = x^4$. This implies that along the x-axis, the function appears to have a minimum at $x = 0$.
- $f(0, y) = y^4$. This implies that along the y-axis, the function also appears to have a minimum at $y = 0$.
- $f(x, x) = 2x^4 - x^4 = x^4$. This implies that along the line $x = y$, the function appears to have a minimum at $(x, y) = (0, 0)$.
- $f(x, -x) = 2x^4 - x^4 = x^4$, and again we see we get a minimum in this direction.

It is important to state: this does not generally imply we have a minimum overall! It is possible that the contour lines going through the saddle could 'squeeze in' between some of these lines, giving a maximum in some direction we haven't yet tested.

Fortunately though, in this case symmetry of the function saves us. We have symmetry in the lines $x = y$ and $y = -x$, which tells us that the contour lines for a saddle would also have to have this symmetry. But then they would have to include the lines $x = y$ and $y = -x$ in the different regions of 'maximum direction' and 'minimum direction', so this cannot be a saddle. We conclude that this must be a minimum. Sketching the contours, we have:



In general, we don't know the exact shape that has been given in the above plot - but the polar argument will confirm it.

(c) This function can also be expressed in polar coordinates as:

$$f(r,\theta) = r^4\cos^4(\theta) + r^4\sin^4(\theta) - r^4\sin^2(\theta)\cos^2(\theta) = r^4\left(\cos^4(\theta) + \sin^4(\theta) - \sin^2(\theta)\cos^2(\theta)\right).$$

Now observe that:

$$\cos^4(\theta) = \frac{1}{2^4}(e^{i\theta} + e^{-i\theta})^4 = \frac{1}{8}\left(\cos(4\theta) + 4\cos(2\theta) + 3\right).$$

Furthermore, we have:

$$\begin{aligned}
\sin^4(\theta) - \sin^2(\theta)\cos^2(\theta) &= \sin^2(\theta)(\sin^2(\theta) - \cos^2(\theta)) \\
&= -\frac{1}{2}\cos(2\theta)\left(1 - \cos(2\theta)\right) \\
&= \frac{1}{2}\cos^2(2\theta) - \frac{1}{2}\cos(2\theta) \\
&= \frac{1}{4}(1 + \cos(4\theta)) - \frac{1}{2}\cos(2\theta).
\end{aligned}$$

Putting things together, we see that:

$$\cos^4(\theta) + \sin^4(\theta) - \sin^2(\theta)\cos^2(\theta) = \frac{5}{8} + \frac{3}{8}\cos(4\theta) = \frac{1}{8}\left(3\cos(4\theta) + 5\right),$$

giving the required form.

Now, r^4 is certainly increasing, and the angular term oscillates between $2 \leq 3\cos(4\theta) + 5 \leq 8$. Hence the function is everywhere positive apart from $(0,0)$, with the function values coming down slightly when $\theta = \frac{1}{4}(2n+1)\pi$, and the function values coming up slightly when $\theta = \frac{1}{2}n\pi$. This means that contour lines come in a bit closer to the origin when $\theta = \frac{1}{2}n\pi$ (the function reaches a particular value faster in this direction), and a bit further when $\theta = \frac{1}{4}(2n+1)\pi$. This agrees with the plot we produced above.

7. (**Another polar example**) Consider the function:

$$f(x,y) = (xy - y)e^{2x-x^2-y^2}.$$

By writing $x - 1 = r\cos(\theta), y = r\sin(\theta)$, express the function in polar coordinates about $(1,0)$. Hence show that the function has five stationary points, classify them, and produce the contour plot for the function. How does this analysis compare with the second derivative approach?

Solution: Following the advice of the question, we note that the function can be written as:

$$(xy - y)e^{2x-x^2-y^2} = (x-1)ye^{1-(x-1)^2-y^2} = r^2e^{1-r^2}\cos(\theta)\sin(\theta) = \frac{1}{2}\sin(2\theta)r^2e^{1-r^2}.$$

Extremising in the radial direction, we have:

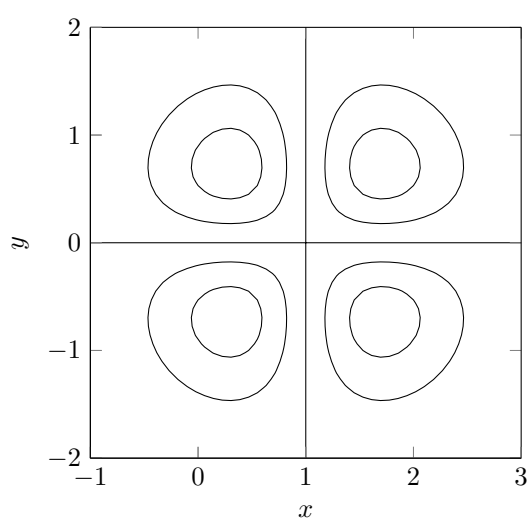
$$\frac{d}{dr}\left(r^2e^{-r^2}\right) = \left(2r - 2r^3\right)e^{-r^2} = 0$$

when $r = 0$ or $r = 1$ (we ignore $r = -1$, which is not an allowed valued). Hence the radial part of the function is increasing initially (driven by the r^2 term, which is most important near $r = 0$) before the radial part is decreasing for larger r (driven by the e^{-r^2} term, which is more important for large values of r).

In the θ direction, note that $\sin(2\theta)$ is maximised at $\theta = \pi/4, -3\pi/4$. Similarly $\sin(2\theta)$ is minimised when $\theta = 3\pi/4, -\pi/4$. Along these directions, we see that in the angular direction we have achieved maxima/minima, and similarly in the radial direction at $r = 1$ we achieve maxima/minima. Near the origin $r = 0$, the angular directions give the function increasing/decreasing respectively depending on the sign of sine. Thus:

- $r = 0$ is a saddle, corresponding to the point $(1,0)$.
- $r = 1$ with $\theta = \pi/4, -3\pi/4$ are maxima, corresponding to the points:
$$\left(1 \pm \frac{1}{\sqrt{2}}, 1 \pm \frac{1}{\sqrt{2}}\right).$$
- $r = 1$ with $\theta = -\pi/4, 3\pi/4$ are minima, corresponding to the points:
$$\left(1 \pm \frac{1}{\sqrt{2}}, 1 \mp \frac{1}{\sqrt{2}}\right).$$

Putting all this information together, we can produce the contour plot, given below.



8. In light of Questions 2-7, make a list of things you could consider when producing a contour plot of a function of two variables $f(x, y)$.

Solution: Using everything we have learned in Questions 2-7, a good list might be:

- Consider taking the gradient of the function to determine the locations of the stationary points.
- Use the second derivative test to determine whether stationary points are maxima, minima, or saddles.
- Compute the contour lines through saddles, using a quadratic approximation, to work out roughly where the contours move out from a saddle.
- Consider the large x, y behaviour of the function to examine what happens to contours as $x \to \infty, y \to \infty$, or as $x^2 + y^2 \to \infty$ (the distance from the origin becomes large).
- Consider symmetries of the function, such as $x \leftrightarrow -x$ (reflection in the line $x = 0$), $y \leftrightarrow -y$ (reflection in the line $y = 0$), $x \leftrightarrow y$ (reflection in the line $y = x$) and $x \leftrightarrow -y$ (reflection in the line $y = -x$). This can reduce the amount of plotting to a small quadrant, and avoid us having to compute the nature of multiple stationary points.
- Calculate the zero contours, and hence determine regions of positivity and negativity of a function. This can help classify stationary points which are surrounded by zero directions.
- Calculate analytical equations of contours, where possible. Usually these will be circular, elliptical, or hyperbolic.
- Use reasoning based on positivity and negativity about a point to deal with cases where the second derivative test fails, for example.
- Consider the function along the axes $x = 0, y = 0$ and the lines $x = y, x = -y$ to reduce the problem to a one-dimensional problem along these directions. This can aid the classification of certain stationary points if they lie on these lines.
- Switch to polar coordinates, attempting to write the function in the form $f(r, \theta) = h(r)g(\theta)$. This allows us to reason about the radial profile and angular profile of the function separately.
- Consider the allowed domain of the function; for example, if $\log(x)$ is involved, we must have $x > 0$.

9. Produce contour plots for each of the following functions, in each case: (i) indicating the nature and location of the stationary points; (ii) providing equations for the contour lines through saddles; and (iii) adding some arrows to your sketch showing the direction of the gradient vector ∇f at different points.

(a) $x^4 + y^4 - 36xy$;

(b) $4x^2 + 4y^2 + x^4 - 6x^2y^2 + y^4$;

(c) $2x^3 + 6xy^2 - 3y^3 - 150x$;

(d) $\frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$;

(e) $xy(x^2 + y^2 - 1)$;

(f) $x^6 + y^6 - 24x^2y^2$;

(g) $x\ln(x) - y^2$;

(h) $(1 - x^2 - y^2)^2 + \frac{7}{2}\ln(3 + y^2)$;

(i) $\dfrac{1}{x^2 + y^2 + 1}$;

(j) $\dfrac{2(y - x^2)}{1 + x^2 + y^2}$;

(k) $(x^2 - y^2)e^{-x^2-y^2}$;

(l) $(1 - 4xy)\exp(-x^2 - y^2)$;

(m) $e^x + e^y - x - y$;

(n) $e^x - e^y + x - y$;

(o) $\exp(\frac{1}{2}(x^2 + y^2) + \frac{1}{4}x^4)$;

(p) $\exp(\frac{1}{2}(x^2 + y^2) - \frac{1}{4}x^4)$;

(q) $\sin(x)\sin(y)$;

(r) $1 - \cos(x) + \frac{1}{2}y^2$.

Solution:

(a) For the function $f(x, y) = x^4 + y^4 - 36xy$, we know for large x, y the contours will begin to resemble $x^4 + y^4 \approx k$, which look like 'squared off circles'. Taking the gradient we have:

$$\nabla(x^4 + y^4 - 36xy) = \big(4x^3 - 36y, 4y^3 - 36x\big) = (0, 0)$$

at stationary points. Multiplying the first equation by x, and the second by y, we obtain:

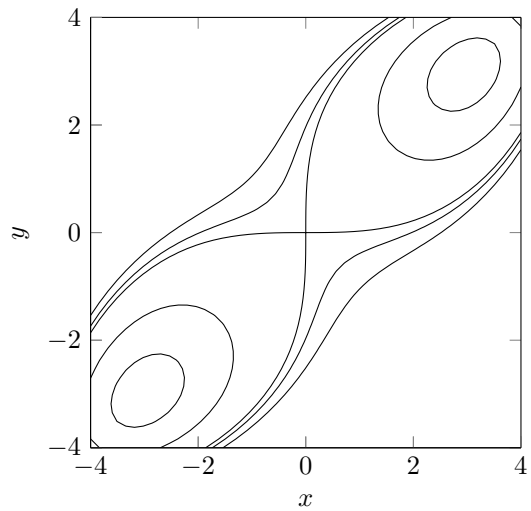
$$\begin{aligned} 4x^4 - 36xy &= 0, \\ 4y^4 - 36xy &= 0. \end{aligned}$$

Subtracting, we see that $x^4 = y^4$, so $x = \pm y$ (considering only real solutions). If $x = y$, then $4x^3 - 36x = 0$, giving $x = 0$ or $x = \pm 3$. If $x = -y$, then $4x^3 + 36x = 0$, giving $x = 0$ or $x^2 = -9$, which cannot occur for real solutions. Thus all stationary points are: $(0, 0)$, $(3, 3)$ and $(-3, -3)$.

Next, we calculate the second derivatives to determine the nature of thee stationary points. We have $f_{xx} = 12x^2$, $f_{xy} = -36$ and $f_{yy} = 12y^2$. Now:

- At $(0, 0)$, we have $f_{xx}f_{yy} - f_{xy}^2 < 0$, so this point is a saddle. Close to the origin, the function becomes $f(x, y) \approx -36xy$, which shows that the contour lines through the saddle are $x = 0$ and $y = 0$.
- At $(3, 3)$, we have $f_{xx}f_{yy} - f_{xy}^2 = 12^2(3^2 \cdot 3^2 - 3^2) > 0$, and $f_{xx} + f_{yy} > 0$. Hence this point is a minimum. Similarly at $(-3, -3)$, we have the same results, implying this point is also a minimum.

We now have everything we need to sketch the contour plot, given on the following page.



(b) For the function $f(x, y) = 4x^2 + 4y^2 + x^4 - 6x^2y^2 + y^4$, we have:

$$\nabla f = (8x + 4x^3 - 12xy^2, 8y - 12x^2y + 4y^3) = (0, 0)$$

at stationary points. Multiplying the first equation by x, and the second by y, we have:

$$\begin{aligned} 8xy + 4x^3y - 12xy^3 &= 0, \\ 8xy - 12x^3y + 4xy^3 &= 0. \end{aligned}$$

Subtracting these equations, we have $4(x^3y - xy^3) + 12(x^3y - xy^3) = 0$, which implies $xy(x^2 - y^2) = 0$. We now break down into cases:

- If $x = 0$, then the second equation implies that $8y + 4y^3 = 0$, giving $y = 0$ or $y^2 = -2$. Hence $(0, 0)$ is the only result.
- Similarly if $y = 0$ then the first equation implies that $x = 0$ or $x^2 = -2$, giving $(0, 0)$ as the only result.
- If $x = y$, then the first equation implies that $8x + 4x^3 - 12x^3 = 0$, hence either $x = 0$ (putting us back in the first case) or $x^2 = 1$, so that $x = \pm 1$. This gives stationary points $(\pm 1, \pm 1)$.
- In the final case, if $x = -y$, the first equation implies that $8x + 4x^3 - 12x^3 = 0$ similarly, giving $(\pm 1, \mp 1)$ as stationary points.

Thus there are five stationary points in total. Note that there is also a high degree of symmetry of this function, however. We have symmetry in the line $y = 0$, since $x \mapsto -x$ is a symmetry. We also have symmetry in the line $x = 0$, since $y \mapsto -y$ is a symmetry. Hence we need only classify the points $(0, 0)$ and $(1, 1)$ (the behaviour at all the other points $(-1, -1)$, $(-1, 1)$ and $(1, -1)$ can be determined by symmetry).

To determine their nature, we compute the second derivatives. We have $f_{xx} = 8 + 12x^2 - 12y^2$, $f_{yy} = 8 - 12x^2 + 12y^2$ and $f_{xy} = -24xy$. Hence:

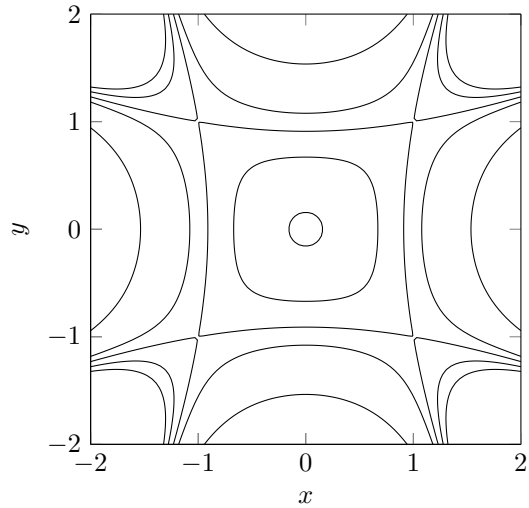
- At $(0, 0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = 64 > 0$, and $f_{xx} + f_{yy} = 16 > 0$, which implies that this point is a minimum.
- At $(1, 1)$, we have $f_{xx}f_{yy} - f_{xy}^2 = 64 - (24)^2 < 0$, so this is a saddle. Writing $x = 1 + \eta$, $y = 1 + \xi$ near the saddle, and keeping only the quadratic terms, we have:

$$\begin{aligned} f(1 + \eta, 1 + \xi) - f(1, 1) &= \frac{1}{2} f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2} f_{yy}\xi^2 + \cdots \\ &= 4\eta^2 - 24\eta\xi + 4\xi^2 + \cdots . \end{aligned}$$

This gives the equation of the contour lines as $4\eta^2 - 24\eta\xi + 4\xi^2 = 0$, which can be rearranged to give:

$$\xi = (3 \pm 2\sqrt{2})\eta \qquad \Rightarrow \qquad y - 1 = (3 \pm 2\sqrt{2})(x - 1).$$

At this stage, we have enough information to prepare the contour sketch.



(c) For the function $f(x, y) = 2x^3 + 6xy^2 - 3y^3 - 150x$, we have at stationary points:

$$\nabla f = (6x^2 + 6y^2 - 150, 12xy - 9y^2) = (0, 0).$$

This gives the equations:

$$\begin{aligned} 6x^2 + 6y^2 - 150 &= 0, \\ 3y(4x - 3y) &= 0. \end{aligned}$$

Solving the second equation, note that if $y = 0$, we have from the first equation that $x = \pm 5$. Hence $(\pm 5, 0)$ are stationary points.

In the second case, we have $4x = 3y$. Substituting into the first equation, we have:

$$x^2 + y^2 - 25 = 0 \qquad \Rightarrow \qquad \frac{9y^2}{16} + y^2 - 25 = 0 \qquad \Rightarrow \qquad y^2 = 16.$$

Thus $y = \pm 4$, which gives $x = \pm 3$. Thus $(\pm 3, \pm 4)$ are additionally stationary points.

To determine the nature of all the points, we compute the second derivatives. We have $f_{xx} = 12x$, $f_{yy} = 12x - 18y$ and $f_{xy} = 12y$. Hence:

- At $(\pm 5, 0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = 144(\pm 5)^2 > 0$. We also have $f_{xx} + f_{yy} = 24(\pm 5)$, so that point $(5, 0)$ is a minimum and the point $(-5, 0)$ is a maximum.
- At $(3, 4)$, we have $f_{xx}f_{yy} - f_{xy}^2 = (36)(36 - 72) - 48 < 0$, so this point is a saddle. Similarly, we have at $(-3, -4)$, $f_{xx}f_{yy} - f_{xy}^2 = (-36)(-36 + 72) + 48 < 0$, so this point is also a saddle.
  Computing the contour lines, about $(3, 4)$ we set $x = 3 + \eta$, $y = 4 + \xi$. Then keeping only quadratic terms, we have:

$$\begin{aligned} f(3 + \eta, 4 + \xi) - f(3, 4) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= 18\eta^2 + 48\eta\xi - 36\xi^2 + \cdots . \end{aligned}$$

Hence the contour lines are $3\eta^2 + 8\eta\xi - 6\xi^2 = 0$. Rearranging, we see that:

$$\xi = \frac{-4\eta \pm \sqrt{64\eta^2 + 72\eta^2}}{-12} = \left(\frac{2 \mp \sqrt{34}}{6}\right)\eta$$

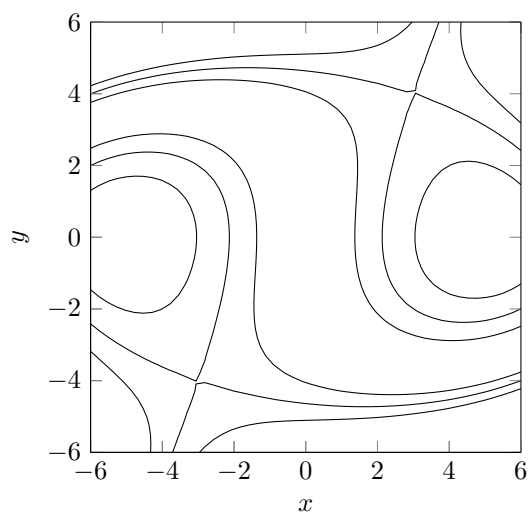
are the equations of the contour lines. One clearly has positive gradient, the other has negative gradient.

Similarly, about $(-3, -4)$, we set $x = -3 + \eta$, $y = -4 + \xi$. Then keeping only quadratic terms, we have:

$$\begin{aligned} f(-3+\eta, -4+\xi) - f(-3,-4) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= -18\eta^2 - 48\eta\xi + 36\xi^2 + \cdots . \end{aligned}$$

Hence the contour lines are the same as those passing through $(3, 4)$.

We now have everything we need to prepare the contour sketch.



(d) For the function $f(x, y) = \frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$, we take the gradient to obtain:

$$\nabla f = \left(x + 2xy, y + x^2 - y^2\right) = (0, 0)$$

at stationary points. This gives the simultaneous equations:

$$\begin{aligned} x(1 + 2y) &= 0, \\ y + x^2 - y^2 &= 0. \end{aligned}$$

The first equation implies that $x = 0$ or $y = -\frac{1}{2}$. If $x = 0$, the second equation implies that $y = 0$ or $y = 1$, so we get stationary points $(0, 0), (0, 1)$. In the second case, if $y = -\frac{1}{2}$, the second equation implies that $x^2 = 3/4$, giving $x = \pm\sqrt{3/4}$. Hence the complete list of stationary points is:

$$(0, 0), \qquad (0, 1), \qquad (\pm\sqrt{3/4}, -1/2).$$

Observe also that the function is symmetrical under $x \leftrightarrow -x$, implying that it is symmetrical about the y-axis.

To classify the points, we consider the second derivatives. We have $f_{xx} = 1 + 2y$, $f_{yy} = 1 - 2y$, $f_{xy} = 2x$. Thus:

- At $(0, 0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = 1 > 0$, and $f_{xx} + f_{yy} = 2 > 0$. Hence this is a minimum.

- At $(0,1)$, we have $f_{xx}f_{yy} - f_{xy}^2 = -3 < 0$, so this is a saddle. Setting $x = \eta, y = 1 + \xi$, we see that near the saddle point (keeping only terms to quadratic order):

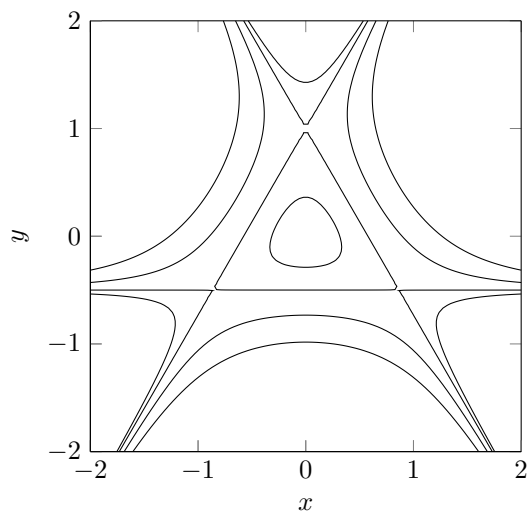
$$\begin{aligned} f(\eta, 1+\xi) - f(0,1) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= \frac{3}{2}\eta^2 - \frac{1}{2}\xi^2 + \cdots . \end{aligned}$$

  It follows that the contour lines passing through the saddles are $\frac{3}{2}\eta^2 = \frac{1}{2}\xi^2$, which can be re-expressed as $\xi = \pm\sqrt{3}\eta$. Rewriting in terms of x, y, the contour lines are $y - 1 = \pm\sqrt{3}x$.
- At $(\sqrt{3/4}, -1/2)$, we have $f_{xx}f_{yy} - f_{xy}^2 = -3 < 0$. Hence this is a saddle. Setting $x = \sqrt{3/4} + \eta$ and $y = -1/2 + \xi$, we see that near the saddle point (keeping only terms to quadratic order):

$$\begin{aligned} f(\sqrt{3/4} + \eta, -1/2 + \xi) - f(\sqrt{3/4}, -1/2) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= 2\sqrt{\frac{3}{4}}\eta\xi + \xi^2 + \cdots . \end{aligned}$$

  So the contour lines passing through this saddle are $\xi = 0$ and $\xi = -\sqrt{3}\eta$. Rewriting in terms of x, y, we have $y = -1/2$ and $y + 1/2 = -\sqrt{3}(x - \sqrt{3/4})$.
- Similarly, by symmetry, $(-\sqrt{3/4}, -1/2)$ must be a saddle, with contour lines $y + 1/2 = \sqrt{3}(x + \sqrt{3/4})$.

We now have everything we need to sketch the contours of the function.



A more efficient way of dealing with this problem, if you can spot it, is to use polar coordinates (which explains the three-fold rotational symmetry the function seems to have in the contour plot above). In polars, the function becomes:

$$f(r,\theta) = \frac{1}{2}r^2 + r^3\left(\cos^2(\theta)\sin(\theta) - \frac{1}{3}\sin^3(\theta)\right) = \frac{1}{2}r^2 + r^3\left(\sin(\theta) - \frac{4}{3}\sin^3(\theta)\right).$$

Using the identity $\sin^3(\theta) = \frac{3}{4}\sin(\theta) - \frac{1}{4}\sin(3\theta)$, we see that the function may be written as:

$$f(r,\theta) = \frac{1}{2}r^2 + \frac{1}{3}r^3\sin(3\theta).$$

This shows the rotational symmetry present in the figure above. We can also calculate stationary points in polar coordinates this way:

$$\frac{\partial f}{\partial r} = r + r^2 \sin(3\theta) = 0, \qquad \frac{\partial f}{\partial \theta} = r^2 \cos(3\theta) = 0.$$

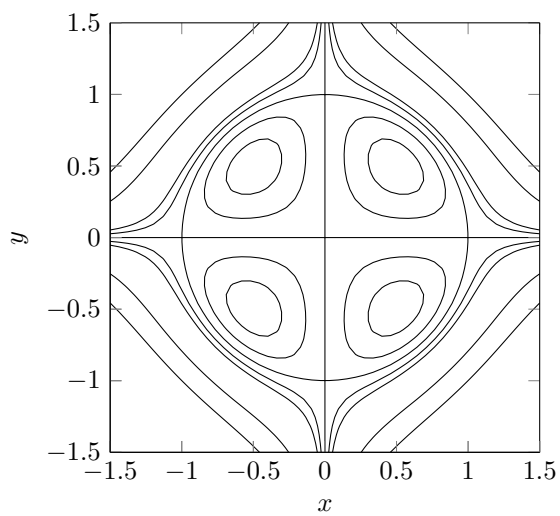
The first equation implies either $r = 0$ (which is a stationary point, and is a minimum as we discovered above), or tells us that $r = -\sin(3\theta)$. Inserting into the second equation, we have $\sin^2(3\theta)\cos(3\theta) = 0$. A zero of $\sin(3\theta)$ just gives us the stationary point we have already found. It remains to have stationary points at $\cos(3\theta) = 0$ where $\sin(3\theta) < 0$. This includes $\theta = \pi/2, \theta = -\pi/6$ and $\theta = -5\pi/6$. These correspond exactly to the three saddles we found above.

(e) For the function $f(x, y) = xy(x^2 + y^2 - 1)$, the use of zero contours seems like a good idea. The zero contours are easily calculated to be $x = 0, y = 0$ and $x^2 + y^2 = 1$. This means that there must be saddles at the points $(0, 0), (\pm 1, 0), (0, \pm 1)$, because zero contours are crossing here.

Alternatively, polar coordinates are a great bet (and can help us determine if we have any additional stationary points). We have $f(r, \theta) = r^2(r^2 - 1)\cos(\theta)\sin(\theta) = \frac{1}{2}r^2(r^2 - 1)\sin(2\theta)$. It follows that the function can be factorised as $f(r, \theta) = h(r)g(\theta)$. The function $h(r) = \frac{1}{2}r^4 - \frac{1}{2}r^2$ can be extremised via:

$$h'(r) = 2r^3 - r = 0,$$

giving stationary points at $r = 0$ or $r = 1/\sqrt{2}$. The origin must correspond to a saddle, as we discussed above, because the angular part $\sin(2\theta)$ is negative in some regions and positive in others. The point $1/\sqrt{2}$ will correspond to maxima in regions where $\sin(2\theta)$ is maximised (at $\theta = \pi/4, -3\pi/4$) and minima in regions where $\sin(2\theta)$ is minimised (at $\theta = 3\pi/4, -\pi/4$). This gives us enough information to complete the sketch:



(f) Note that the function $f(x, y) = x^6 + y^6 - 24x^2y^2$ takes the form $f(x, y) \approx x^6 + y^6$ for large x, y, implying the contours are 'squared-off circles' for sufficiently large (x, y). Note that near the origin, the function approximately takes the form $f(x, y) \approx -24x^2y^2$, implying that the contours through the origin are $x = 0, y = 0$, implying that there is a saddle at the origin (with contour lines $x = 0, y = 0$ passing through the saddle).

There is also a high degree of symmetry for this function, with $x \leftrightarrow -x, y \leftrightarrow -y, x \leftrightarrow y, x \leftrightarrow -y$ all leaving the function invariant. This implies that the contour plot should be invariant under reflections in the lines $x = 0, y = 0, x = y, x = -y$.

Proceeding to calculate the gradient, we have:

$$\nabla f = \left(6x^5 - 48xy^2, 6y^5 - 48x^2y\right) = (0,0)$$

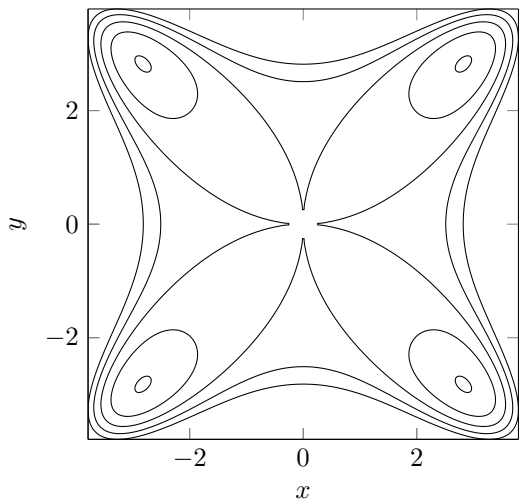
at stationary points. Multiplying the first equation by x, and the second by y, we have $6x^6 - 48x^2y^2 = 0$ and $6y^6 - 48x^2y^2 = 0$, giving $x^6 = y^6$ on comparison. Hence at stationary points, we must have $x = \pm y$.

If $x = y$, the first equation gives $x^5 - 8x^3 = 0$ so that $x = 0$ or $x = \pm 2\sqrt{2}$. Hence $(0,0), \pm(2\sqrt{2}, 2\sqrt{2})$ are stationary points. If $x = -y$, the first equation instead gives $x^5 + 8x^3 = 0$, so that $x = 0$ or $x = -2$. Hence we discover an additional third stationary point at $(-2, 2)$.

To determine the nature of the stationary points, we calculate the second derivatives. We have $f_{xx} = 30x^4 - 48y^2$, $f_{yy} = 30y^4 - 48x^2$ and $f_{xy} = -96xy$. We have already determined that $(0,0)$ is a saddle, and by the symmetries, we need only check $(2\sqrt{2}, 2\sqrt{2})$'s nature. We have:

$$f_{xx}f_{yy} - f_{xy}^2 = (30 \cdot 64 - 48 \cdot 8) \cdot (30 \cdot 64 - 48 \cdot 8) - (-96 \cdot 8)^2 > 0,$$

and $f_{xx} + f_{yy} = (30 \cdot 64 - 48 \cdot 8) + (30 \cdot 64 - 48 \cdot 8) > 0$, so this is a minimum. We now have enough information to prepare the contour sketch.



(g) For the function $f(x,y) = x\ln(x) - y^2$, we notice that we only need sketch in the region $x > 0$, where the function exists. Further, for large values of (x,y), we have that the function gets close to $f(x,y) \approx -y^2$, so the contour lines become roughly horizontal. Computing stationary points, we note that:

$$\nabla f = (\ln(x) + 1, -2y) = (0,0)$$

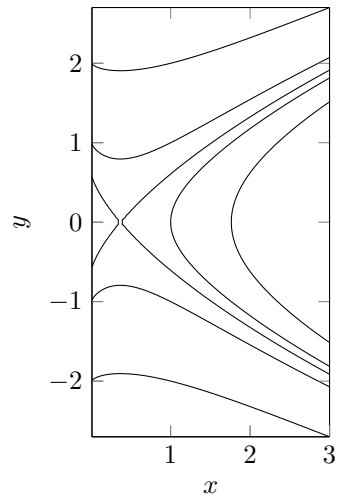
at stationary points. This implies that $\ln(x) = -1$ at stationary points, giving $x = e^{-1}$. Thus the only stationary point is $(1/e, 0)$.

Computing the second derivatives, we have $f_{xx} = 1/x, f_{yy} = -2, f_{xy} = 0$. Thus at the stationary point, we have $f_{xx}f_{yy} - f_{xy}^2 = -2e < 0$, which gives that the stationary point is a saddle. Setting $x = 1/e + \eta, y = \xi$ near the saddle, we have to second order:

$$\begin{aligned} f(1/e + \eta, \xi) - f(1/e, 0) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= \frac{e}{2}\eta^2 - \xi^2 + \cdots, \end{aligned}$$

so the contour lines through the saddle are $\xi = \pm\sqrt{\frac{e}{2}}\eta$. Written in terms of x, y they give $y = \pm\sqrt{\frac{e}{2}}\left(x - \frac{1}{e}\right)$.

We now have all the information needed to perform a sketch.



(h) For the function $f(x,y) = (1-x^2-y^2)^2 + \frac{7}{2}\ln(3+y^2)$, we start by observing that in polar coordinates, we have:

$$f(r,\theta) = (1-r^2)^2 + \frac{7}{2}\ln(3+r^2\sin^2(\theta)).$$

This implies that as $r \to \infty$, we have $f(r,\theta) \approx r^4$, so that the contours become roughly circular in the large radius limit.

This function also has lots of symmetry. It is invariant under $x \leftrightarrow -x$ and $y \leftrightarrow -y$, meaning it is has reflectional symmetry about the lines $x = 0$ and $y = 0$.

Proceeding to calculate the gradient, we have:

$$\nabla f = \left(-4x(1-x^2-y^2), -4y(1-x^2-y^2) + \frac{7y}{3+y^2}\right) = (0,0)$$

at stationary points. The first equation implies that either $x = 0$ or $x^2+y^2 = 1$. Dividing these into cases, we have:

- If $x = 0$, then the second equation implies:

$$-4y(1-y^2) + \frac{7y}{3+y^2} = 0.$$

  Hence either $y = 0$ (giving the stationary point $(0,0)$), or multiplying up, and rearranging, we have:

$$-4(1-y^2)(3+y^2) + 7 = 0 \qquad \Rightarrow \qquad 4y^4 + 8y^2 - 5 = 0 \qquad \Rightarrow \qquad (2y^2-1)(2y^2+5) = 0$$

  The first factor gives $y = \pm 1/\sqrt{2}$. This results in the stationary points $(0, \pm 1/\sqrt{2})$.
- If $x^2+y^2 = 1$, then the second equation implies:

$$\frac{7y}{3+y^2} = 0.$$

  This gives the stationary points $(\pm 1, 0)$.

So overall, there are five stationary points $(0,0)$, $(0,\pm 1/\sqrt{2})$, $(\pm 1, 0)$.

Taking a second derivative, we have:

$$f_{xx} = -4 + 12x^2 + 4y^2, \qquad f_{xy} = 8xy, \qquad f_{yy} = -4 + 4x^2 + 12y^2 + \frac{7(3-y^2)}{(3+y^2)^2}.$$

Hence:

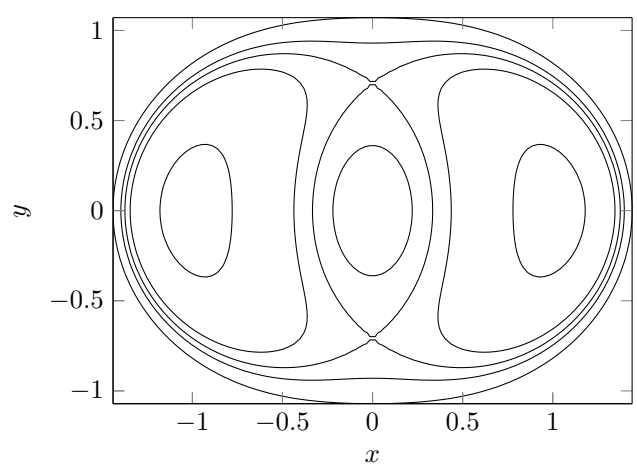
- At $(0,0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = (-4)(-4+7/3) > 0$ and $f_{xx} + f_{xy} = -4 + (-4+7/3) < 0$. Hence this point is a maximum.
- At $(1,0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = (8)(7/3) > 0$, and $f_{xx} + f_{xy} = 8 + 7/3 > 0$. Hence this point is a minimum. Similarly, $(-1,0)$ must be a minimum (by the symmetry $x \leftrightarrow -x$).
- At $(0, 1/\sqrt{2})$, we have $f_{xx}f_{yy} - f_{xy}^2 = (-2)(2+10/7) < 0$. Hence this point is a saddle. Setting $x = \eta$, $y = 1/\sqrt{2} + \xi$, we have to quadratic order:

$$\begin{aligned} f(\eta, 1/\sqrt{2}+\xi) - f(0, 1/\sqrt{2}) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= -\eta^2 + \frac{12}{7}\xi^2 + \cdots . \end{aligned}$$

  Hence the contours through this saddle are $\xi = \pm\sqrt{\frac{7}{12}}\eta$, which written in terms of x, y are $y - 1/\sqrt{2} = \pm\sqrt{\frac{7}{12}}x$.

  Similarly at $(0, -1/\sqrt{2})$, we have a saddle, with contour lines $y + 1/\sqrt{2} = \pm\sqrt{\frac{7}{12}}x$.

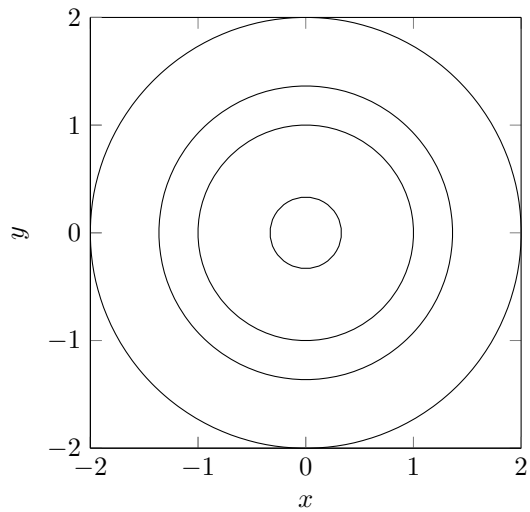
We now have enough information to produce the plot.



(i) For the function $f(x,y) = \dfrac{1}{x^2+y^2+1}$, it is obvious that switching to polars is a good idea. In polars, we have:

$$f(r,\theta) = \frac{1}{r^2+1}.$$

This implies that the contour lines are $r^2 + 1 = k$, giving $r = c$, a constant. Thus the contour lines are exactly circles, centred on the origin. As r increases, f decreases, so evidently the origin $(0,0)$ is a maximum. We can easily prepare the contour sketch as a result:



(j) For the function:

$$f(x,y) = \frac{2(y-x^2)}{1+x^2+y^2},$$

we immediately notice that there is a zero contour at $y = x^2$. The function is positive in the region $y > x^2$, and negative in the region $y < x^2$. Further, there is symmetry under the transformation $x \leftrightarrow -x$, telling us the plot should have reflectional symmetry in the $y = 0$ axis.

Taking the gradient, at a stationary point we have:

$$\nabla f = \left(\frac{-4x(1+x^2+y^2) - 4x(y-x^2)}{(1+x^2+y^2)^2}, \frac{2(1+x^2+y^2) - 4y(y-x^2)}{(1+x^2+y^2)^2} \right) = (0,0).$$

Hence, we have the simultaneous equations:

$$\begin{aligned} x\left(y^2+y+1\right) &= 0 \\ 1+x^2-y^2+2yx^2 &= 0 \end{aligned}$$

The first equation implies $x = 0$, since the other factor has no real roots. This implies $y = \pm 1$, giving only two stationary points at $(0, \pm 1)$.

As $y \to \infty$, we see that $f \to 0$, so the point $(0,1)$ is entirely surrounded on all sides by zeroes, and takes the value 1 there. Hence, it must be a maximum.

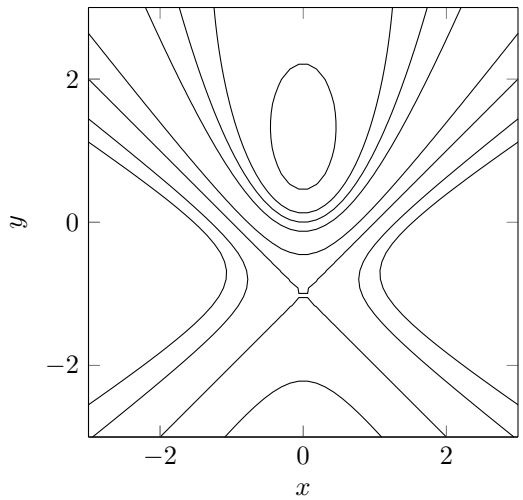
For the stationary point at $(0,-1)$, note that $f(0,-1) = -1$. Furthermore, in this case, we may calculate the analytical equations of the contours through this point, since they take the form:

$$-1 = \frac{2(y-x^2)}{1+x^2+y^2} \qquad \Rightarrow \qquad -1-x^2-y^2 = 2y-2x^2.$$

Rearranging, they can be put in the form:

$$(y+1)^2 = x^2 \qquad \Rightarrow \qquad y+1 = \pm x,$$

which shows that this point is a saddle, and determines the *exact contours* passing through it. This gives all the information we need to sketch.



(k) For the function $f(x, y) = (x^2 - y^2)e^{-x^2-y^2}$, it's a good idea to use polar coordinates. We have:

$$f(r, \theta) = r^2 \left(\cos^2(\theta) - \sin^2(\theta)\right) e^{-r^2} = r^2 e^{-r^2} \cos(2\theta).$$

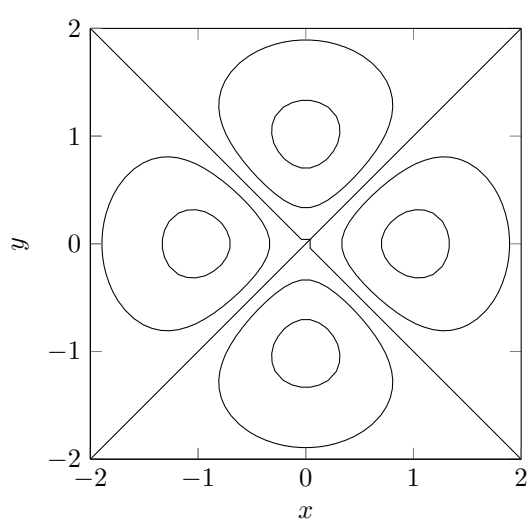
Hence, we see that the function factorises in the form $h(r)g(\theta)$. Analysing the radial profile of the function, we note that:

$$\frac{d}{dr}\left(r^2 e^{-r^2}\right) = \left(2r - 2r^3\right) e^{-r^2},$$

which gives stationary values at $r = 0$ and $r = 1$. Near $r = 0$, the r^2 behaviour dominates the radial profile, so the profile increases before decreasing (at large r it is dominated by e^{-r^2}).

This is modulated by the angular part of the function, $\cos(2\theta)$. At $r = 0$, this angular part ensures that there are regions of positive and negative values of the function near 0, implying it is a saddle. Looking back at the function, the zero contours in Cartesian coordinates are $x = \pm y$, which give the contour lines passing through this saddle.

The angular part is maximised when $2\theta = 0, 2\pi$, giving $\theta = 0, \pi$. Hence $(1, 0), (-1, 0)$ are maxima. The angular part is minimised when $2\theta = \pi, -\pi$, giving $\theta = \pi/2, -\pi/2$. Hence $(0, 1), (0, -1)$ are minima. This gives us enough information to produce the contour plot:



(l) For the function $f(x,y) = (1-4xy)\exp(-x^2-y^2)$, note first that there is symmetry under $x \leftrightarrow y$ and $x \leftrightarrow -y$. Hence the contour plot has reflectional symmetry in the lines $x = y$ and $x = -y$.

We have the gradient:

$$\nabla f = (-4y - 2x(1-4xy), -4x - 2y(1-4xy))\, e^{-x^2-y^2} = (0,0),$$

at stationary points. This implies the simultaneous equations:

$$\begin{aligned} -4y - 2x + 8x^2y &= 0, \\ -4x - 2y + 8xy^2 &= 0. \end{aligned}$$

Multiplying the first equation by y and the second by x, we have:

$$\begin{aligned} 2y^2 + xy - 4x^2y^2 &= 0, \\ 2x^2 + xy - 4x^2y^2 &= 0. \end{aligned}$$

Subtracting the equations, we have $x^2 = y^2$, so that at stationary points we need $x = \pm y$. If $x = y$, we have from the first equation $-6x + 8x^3 = 0$, implying $x = 0$ or $x = \pm\sqrt{3}/2$. Hence we have the stationary points $(0,0)$, $\pm(\sqrt{3}/2, \sqrt{3}/2)$. On the other hand if $x = -y$, we have from the first equation $2x - 8x^3 = 0$, which implies $x = 0$ or $x = \pm 1/2$. Thus we get the additional stationary points $\pm(1/2, -1/2)$.

The zero contours of this function are $1 = 4xy$, giving $y = 1/4x$. For $y > 1/4x$, the function is negative, whilst if $y < 1/4x$, the function is positive. Further, the function must tend to zero at infinity. This implies that $\pm(\sqrt{3}/2, \sqrt{3}/2)$ have to be minima and $\pm(1/2, -1/2)$ have to be maxima, without calculating the second derivatives.

This function also provides an opportunity to consider the lines $x = y$, $x = -y$, reducing the problem to the one dimensional case. Along $x = y$, the function is (near $x = 0$):

$$f(x,x) = (1-4x^2)e^{-2x^2} = (1-4x^2)(1-2x^2+\cdots) = 1 - 6x^2 + \cdots$$

This implies that the function appears to have a *maximum* in this direction at $(0,0)$. On the other hand, in the direction $x = -y$, we have (near $x = 0$):

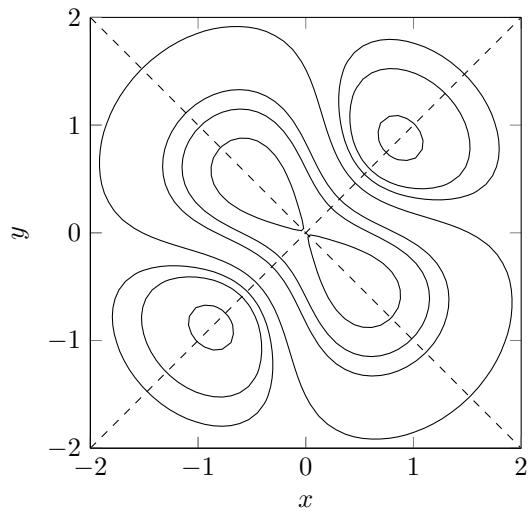
$$f(x,-x) = (1+4x^2)e^{-2x^2} = (1+4x^2)(1-2x^2+\cdots) = 1 + 2x^2 + \cdots$$

which implies that the function appears to have a *minimum* in this direction at $(0,0)$. Thus $(0,0)$ must be a saddle.

We also have for small (x,y):

$$f(x,y) - f(0,0) \approx -x^2 - 4xy - y^2 + \cdots,$$

which implies that the contour lines are $y^2 + 4xy + x^2 = 0$, which can be rearranged to $y = (-2 \pm \sqrt{3})x$. This gives us all the information we need to sketch the contours.

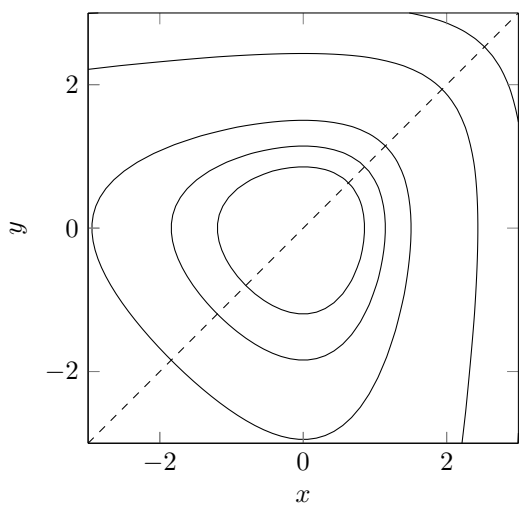


(m) The function $f(x, y) = e^x + e^y - x - y$ has gradient:

$$\nabla f = (e^x - 1, e^y - 1)\,,$$

giving the stationary point $(x, y) = (0, 0)$. Taking the second derivative, we have $f_{xx} = e^x, f_{yy} = e^y, f_{xy} = 0$, which implies that $f_{xx}f_{yy} - f_{xy}^2 = 1 > 0$ and $f_{xx} + f_{yy} > 0$ at $(0, 0)$. Thus $(0, 0)$ is a maximum (so contour lines near $(0, 0)$ look roughly circular).

Note we also have symmetry $x \leftrightarrow y$, so there is symmetry in the line $x = y$. We also have for large y that the function takes the form $e^y - y$, so the contours lines become approximately $e^y - y \approx k$. Thus they become approximately horizontal. Similarly, for large x the contour lines become approximately vertical.



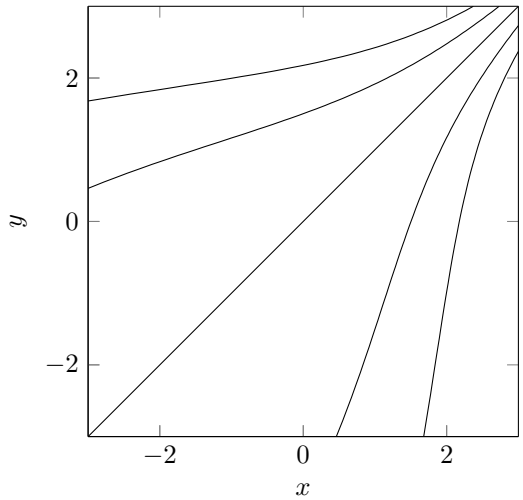
(n) The function $f(x, y) = e^x - e^y + x - y$ has gradient:

$$\nabla f = (e^x + 1, -e^y - 1)\,,$$

which has no stationary points, since $e^x \neq -1$ and $e^y \neq -1$. Uh oh!

We observe by inspection that there is a zero contour at $x = y$. If $y > x$, then the function is negative, and if $x < y$, the function is positive.

We also observe that for large y, the function takes the form $-e^y - y$. So the contour lines are roughly of the form $-e^y - y \approx k$, i.e. they become approximately horizontal. The same happens for large x, the contour lines become approximately horizontal. This is enough information to give a contour plot, without any stationary points!

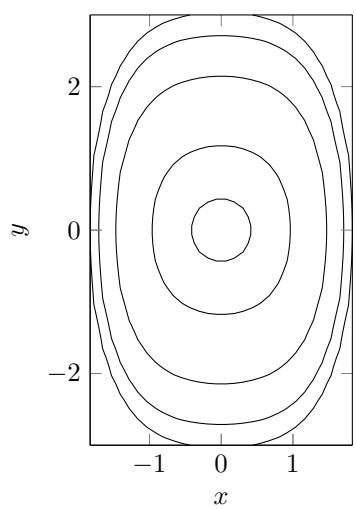


(o) For the function $f(x, y) = \exp(\frac{1}{2}(x^2 + y^2) + \frac{1}{4}x^4)$, we note that for large x, the contours look like $e^{x^4/4} \approx k$, which means they are roughly vertical. We also note that for x, y close to zero, the contours look like $e^{r^2/2} \approx k$ in polar coordinates, so they are roughly circular near the origin.

The gradient of the function is:

$$\nabla f = \left(x + x^3, y\right) \exp\left(\frac{1}{2}(x^2 + y^2) + \frac{1}{4}x^4\right) = (0, 0).$$

Hence the only stationary point occurs when $x = 0, y = 0$ (since $1 + x^2 \neq 0$ for x real). We have seen above that this is a minimum. Hence, we have enough information to sketch the contour plot.



(p) For the function $f(x, y) = \exp(\frac{1}{2}(x^2+y^2)-\frac{1}{4}x^4)$, very similar to the previous part, the contours look roughly vertical for large x, and circular for small x, y. Note this function is also symmetric under $x \leftrightarrow -x$, so there is reflectional symmetry in the y-axis.

The gradient of the function is:

$$\nabla f = \left(x - x^3, y\right) \exp\left(\frac{1}{2}(x^2 + y^2) - \frac{1}{4}x^4\right) = (0, 0).$$

Hence the stationary points are $(x, y) = (0, 0), (1, 0), (-1, 0)$. The second derivatives are:

$$f_{xx} = \left((x - x^3)^2 + 1 - 3x^2\right) f, \qquad f_{xy} = (x - x^3)yf, \qquad f_{yy} = \left(1 + y^2\right) f.$$

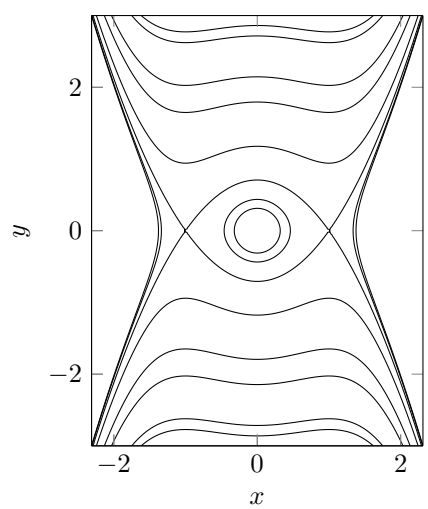
Hence:

- At $(0, 0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = 1 > 0$, and $f_{xx} + f_{yy} = 2 > 0$, so this point is a minimum.
- At $(1, 0)$, we have $f_{xx}f_{yy} - f_{xy}^2 = -2f^2 < 0$, so this point is a saddle. Near $(1, 0)$, we may write $x = 1 + \eta$, $y = \xi$, giving the expansion to quadratic order:

$$\begin{aligned} f(x, y) - f(1, 0) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= e^{1/4}\left(-\eta^2 + \frac{1}{2}\xi^2\right) + \cdots, \end{aligned}$$

  hence the equations of the contour lines through $(1, 0)$ are $\xi = \pm\sqrt{2}\eta$. This gives $y = \pm\sqrt{2}(x - 1)$.
- By reflectional symmetry, $(-1, 0)$ is also a saddle. The equations of the contour lines are, by symmetry, $y = \pm\sqrt{2}(x + 1)$.

We now have enough information to sketch the contour plot.



(q) For the function $f(x, y) = \sin(x)\sin(y)$, we note that:

$$\nabla f = (\cos(x)\sin(y), \sin(x)\cos(y)).$$

Requiring simultaneously that these both be zero implies $\cos(x) = 0$ and $\cos(y) = 0$, or alternatively $\sin(y) = 0$ and $\sin(x) = 0$ (this is because cosine and sine cannot simultaneously be zero). Therefore, the stationary points are $((n + \frac{1}{2})\pi, (m + \frac{1}{2})\pi)$ and $(n\pi, m\pi)$ for all integers n, m.

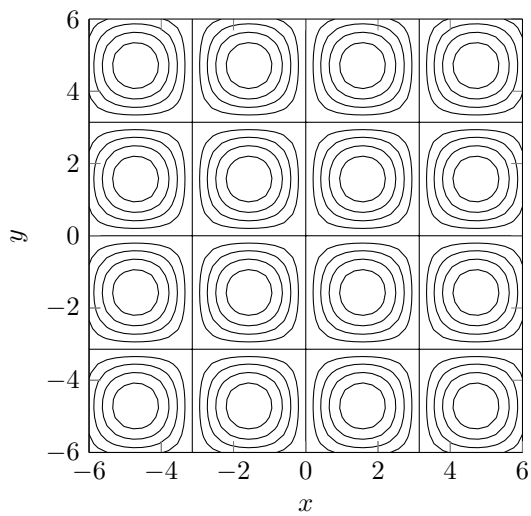
Observe that $f_{xx} = -\sin(x)\sin(y)$, $f_{yy} = -\sin(x)\sin(y)$ and $f_{xy} = \cos(x)\cos(y)$. Hence:

$$f_{xx}f_{yy} - f_{xy}^2 = \sin^2(x)\sin^2(y) - \cos^2(x)\cos^2(y), \qquad f_{xx} + f_{yy} = -2\sin(x)\sin(y).$$

This implies:

- At $((n+\frac{1}{2})\pi, (m+\frac{1}{2})\pi)$, we have $f_{xx}f_{yy} - f_{xy}^2 > 0$, and $f_{xx} + f_{yy} = -2(-1)^{n+m}$. Hence if $n+m$ is even, we have a maximum, and if $n+m$ is odd, we have a minimum.
- At $(n\pi, m\pi)$, we have $f_{xx}f_{yy} - f_{xy}^2 < 0$, so these correspond to saddle points. Along these lines, we also have zero contours of the original function, giving the contour lines through the saddles.

We now have enough information to sketch the contours.



(r) For the function $f(x,y) = 1 - \cos(x) + \frac{1}{2}y^2$, we have:

$$\nabla f = (\sin(x), y) = (0,0)$$

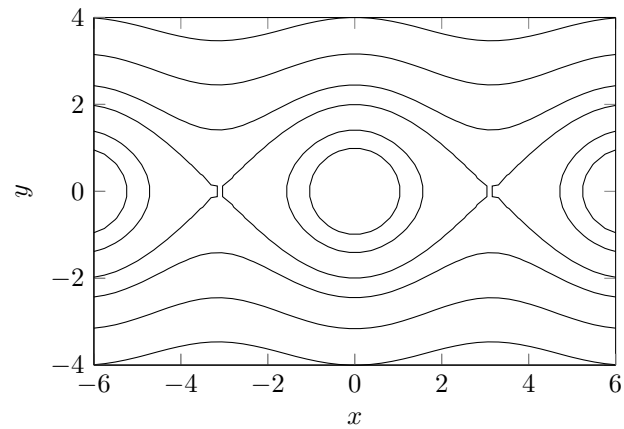
at stationary points. Hence $(x,y) = (n\pi, 0)$ are all the stationary points, for n an integer. Taking the second derivatives, we have $f_{xx} = \cos(x)$, $f_{xy} = 0$ and $f_{yy} = 1$. Hence $f_{xx}f_{yy} - f_{xy}^2 = \cos(x)$. This is positive for n even, and negative for n odd. Hence:

- If n is odd, then $(n\pi, 0)$ is a saddle. Near $(n\pi, 0)$, we may write $x = n\pi + \eta$, $y = \xi$, giving to quadratic order:

$$\begin{aligned} f(n\pi+\eta, \xi) - f(n\pi, 0) &= \frac{1}{2}f_{xx}\eta^2 + f_{xy}\eta\xi + \frac{1}{2}f_{yy}\xi^2 + \cdots \\ &= -\frac{1}{2}\eta^2 + \frac{1}{2}\xi^2 + \cdots, \end{aligned}$$

  which shows that the contour lines through these points are $\eta = \pm\xi$, i.e. $x - n\pi = \pm y$. These are at right-angles to one another.
- If n is even, then $f_{xx} + f_{yy} = \cos(n\pi) + 1 = 2 > 0$, hence these points are minima.

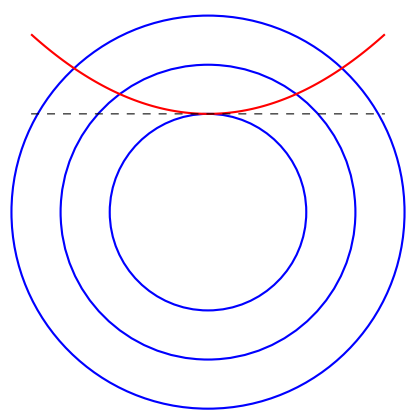
For large y, the contours become roughly $y \approx$ constant, i.e. horizontal lines. This gives us everything we need to sketch the contour plot.



(†) Constrained optimisation using Lagrange multipliers

10. Explain the rationale behind the method of Lagrange multipliers for extremising a function $f(x, y)$ subject to a constraint $g(x, y) = 0$. How does the method of Lagrange multipliers generalise to extremising a function $f(x_1, ..., x_n)$ subject to multiple constraints $g_1(x_1, ..., x_n) = 0$, $g_2(x_1, ..., x_n) = 0$, etc?

Solution: In two dimensions, imagine plotting the contours of the function f and the constraint $g(x, y) = 0$. Below we show the contours of f in blue, and the constraint in red.



We are only allowed to consider points along $g(x, y) = 0$ as part of our extremisation. Assuming we have a neighbouring unconditional maximum (demonstrated by the centre of the circles above), we cannot actually access this, but the next best thing is to use a *point of tangency* between the locus $g(x, y) = 0$ and a contour of constant f. Since ∇g, ∇f are both orthogonal to the contours, we must have that they are *parallel* at points of tangency:

$$\nabla f = \lambda \nabla g \qquad \Rightarrow \qquad \nabla(f - \lambda g) = 0.$$

This shows that maximising the function:

$$L(x, y, \lambda) = f - \lambda g$$

with respect to x, y *and* λ results in the conditional maximum. All this applies in the case of minima and saddles too.

To extremise a function $f(x_1, ..., x_n)$ subject to multiple constraints $g_1, g_2, ...$, we extremise the function:

$$L(x_1, ..., x_n, \lambda_1, \lambda_2, ...) = f(x_1, ..., x_n) - \lambda_1 g(x_1, ..., x_n) - \lambda_2 g(x_1, ..., x_n) - ...$$

instead.

11. Find the extremal values of the functions: (a) $f_1(x,y) = xy^2$; (b) $f_2(x,y) = e^{-xy}$, subject to the constraint that $x^2+y^2=1$. Now, verify that your answers are correct by solving the constraint by setting $x=\cos(\theta), y=\sin(\theta)$.

Solution:

(a) The Lagrangian is $L(x,y,\lambda) = xy^2 + \lambda(x^2+y^2-1)$. Extremising, we have:

$$\frac{\partial L}{\partial x} = y^2 + 2\lambda x = 0, \qquad \frac{\partial L}{\partial y} = 2xy + 2\lambda y = 0, \qquad \frac{\partial L}{\partial \lambda} = x^2+y^2-1 = 0.$$

Multiplying the first equation by y, and the second by x, we have $y^3+2\lambda yx = 0$, and $2x^2y+2\lambda xy = 0$. This implies that $y^3 = 2x^2y$, implying either $y=0$ or $y^2 = 2x^2$.

If $y=0$, the constraint implies that $x=\pm 1$ and $\lambda = 0$. This gives the extremal points $(\pm 1, 0)$.

If $y^2 = 2x^2$, we can substitute back into the constraint to give $3x^2-1=0$, and hence $x = \pm 1/\sqrt{3}$. the corresponding y values are $y = \pm\sqrt{2/3}$ (finally, we also have $\lambda = -x$). Thus we have extreme points:

$$\left(\pm\frac{1}{\sqrt{3}}, \pm\sqrt{\frac{2}{3}}\right), \qquad \left(\pm\frac{1}{\sqrt{3}}, \mp\sqrt{\frac{2}{3}}\right).$$

(b) The Lagrangian is $L(x,y,\lambda) = e^{-xy} + \lambda(x^2+y^2-1)$. Extremising, we have:

$$\frac{\partial L}{\partial x} = -ye^{-xy} + 2\lambda x = 0, \qquad \frac{\partial L}{\partial y} = -xe^{-xy} + 2\lambda y = 0, \qquad \frac{\partial L}{\partial \lambda} = x^2+y^2-1 = 0.$$

Adding the first two equations, we see that $2\lambda(x+y) = (x+y)e^{-xy}$, which tells us that either $x=-y$ or $2\lambda = e^{-xy}$. Substituting the first case into the constraint give $(x,y) = (\pm 1/\sqrt{2}, \mp 1/\sqrt{2})$ as extreme points.

The second case gives $-ye^{-xy} + xe^{-xy} = 0$ in the first equation, implying that $x=y$. Thus substituting into the constraint, we have $(x,y) = (\pm 1/\sqrt{2}, \pm 1/\sqrt{2})$ as extreme points.

We can also solve the constraint $x^2+y^2=1$ by setting $x=\cos(\theta), y=\sin(\theta)$, which means that we can extremise $f_1(\theta) = \cos(\theta)\sin^2(\theta)$ and $f_2(\theta) = e^{-\cos(\theta)\sin(\theta)}$ without constraint.

- In the first case, $f_1'(\theta) = -\sin^3(\theta) + 2\sin(\theta)\cos^2(\theta) = \sin(\theta)(2\cos^2(\theta) - \sin^2(\theta))$. The first factor gives $\theta = n\pi$, which corresponds to $(\pm 1, 0)$, which are the extreme points we found above. The second factor gives:

$$0 = \sin^2(\theta) - 2\cos^2(\theta) = 3\sin^2(\theta) - 2 \qquad \sin(\theta) = \pm\sqrt{2/3}.$$

  This gives rise to the points $(\pm\sqrt{1/3}, \pm\sqrt{2/3})$ and $(\mp\sqrt{1/3}, \pm\sqrt{2/3})$ we also found above.

- In the second case, $f_2(\theta) = e^{-\frac{1}{2}\sin(2\theta)}$, so the extreme values must be when $\sin(2\theta)$ is maximised and minimised. This occurs when $\theta = \frac{1}{4}(2n+1)\pi$, corresponding to $\cos(\theta) = \pm 1/\sqrt{2}$ and $\sin(\theta) = \pm 1/\sqrt{2}$. This gives rise to all the extreme values we found earlier.

12. A particle is contained inside a box which has orthogonal sides of length a, b and c. The particle's energy E is:

$$E = A\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right),$$

where A is a positive constant. Determine the shape of the box which minimises E, subject to the constraint that the volume of the box V is constant by: (a) using the method of Lagrange multipliers; (b) solving the constraint. Verify that your answers agree.

Solution:

(a) The Lagrangian is:

$$L(a, b, c, \lambda) = A\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right) + \lambda\left(V - abc\right).$$

Extremising, we have:

$$\frac{\partial L}{\partial a} = -\frac{2A}{a^3} - \lambda bc = 0, \qquad \frac{\partial L}{\partial b} = -\frac{2A}{b^3} - \lambda ac = 0, \qquad \frac{\partial L}{\partial c} = -\frac{2A}{c^3} - \lambda ab = 0, \qquad \frac{\partial L}{\partial \lambda} = V - abc = 0.$$

Multiplying the first equation by a, the second by b, and the third by c, we have:

$$\frac{2A}{a^2} = -\lambda abc, \qquad \frac{2A}{b^2} = -\lambda abc, \qquad \frac{2A}{c^2} = -\lambda abc.$$

Hence we must have $1/a^2 = 1/b^2 = 1/c^2$ at the optimal box size. This implies that a, b, c are all equal, because they are all positive if they are sides of a box. This means that the box is a *cube* in this case.

(b) The constraint is $V = abc$, so solving, we have $V/bc = a$. Substituting this into the energy which we wish to minimise, we have:

$$E(b, c) = A\left(\frac{b^2c^2}{V^2} + \frac{1}{b^2} + \frac{1}{c^2}\right).$$

Extremising, we have:

$$\frac{\partial E}{\partial b} = A\left(\frac{2bc^2}{V^2} - \frac{2}{b^3}\right) = 0, \qquad \frac{\partial E}{\partial c} = A\left(\frac{2cb^2}{V^2} - \frac{2}{c^3}\right) = 0.$$

From the first equation, we have:

$$\frac{bc^2}{V^2} = \frac{1}{b^2} \qquad \Rightarrow \qquad b^4c^2 = V^2.$$

Similarly from the second equation, we have $c^4b^2 = V^2$. Comparing these equations, we see that $b^4c^2 = c^4b^2$. Assuming that $b, c \neq 0$ (which must be the case for non-zero volume V), we see that $b = c$. Substituting back into the equation $b^4c^2 = V^2$, we see that $b^6 = V^2$, giving $b = c = V^{1/3}$. Substituting back into $V/bc = a$, we see that $a = V^{1/3}$, so again we get a cube.

13. Show that the largest possible volume of a cuboid inscribed in an ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

is $8abc/\sqrt{27}$.

Solution: Let one of the vertices of the cuboid be (x, y, z). Then by symmetry, the remaining vertices must be at $(\pm x, \pm y, \pm z)$, where we choose the minus signs as appropriate in each slot. The total volume is therefore $(2x)(2y)(2z) = 8xyz$. The Lagrangian therefore is:

$$L(x, y, z, \lambda) = 8xyz + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right).$$

Extremising, we have:

$$\frac{\partial L}{\partial x} = 8yz + \frac{2\lambda x}{a^2} = 0, \qquad \frac{\partial L}{\partial y} = 8xz + \frac{2\lambda y}{b^2} = 0, \qquad \frac{\partial L}{\partial z} = 8xy + \frac{2\lambda z}{c^2} = 0, \qquad \frac{\partial L}{\partial \lambda} = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0.$$

Multiplying the first equation by x, the second by y, and the third by z, we see that:

$$\frac{2\lambda x^2}{a^2} = -8xyz, \qquad \frac{2\lambda y^2}{b^2} = -8xyz, \qquad \frac{2\lambda z^2}{c^2} = -8xyz.$$

Equating all of these, we see that:

$$\frac{\lambda x^2}{a^2} = \frac{\lambda y^2}{b^2} = \frac{\lambda z^2}{c^2}.$$

If $\lambda = 0$, then going back to the original equations, we see that $yz = xy = xz = 0$, which implies all parameters are zero, making it impossible to satisfy the constraint. Hence $\lambda \neq 0$, giving $x^2/a^2 = y^2/b^2 = z^2/c^2$. Substituting into the constraint, we see that $x^2/a^2 = y^2/b^2 = z^2/c^2 = 1/3$. Hence, we have:

$$x = \frac{a}{\sqrt{3}}, \qquad y = \frac{b}{\sqrt{3}}, \qquad z = \frac{c}{\sqrt{3}}.$$

Thus the volume is indeed maximised at:

$$V = \frac{8abc}{\sqrt{27}},$$

as required.

14. The area A of a triangle is given by *Heron's formula* as:

$$A = \sqrt{s(s-a)(s-b)(s-c)},$$

where $s = \frac{1}{2}(a+b+c)$ is the semi-perimeter. Using this formula, show that for a fixed perimeter, the triangle of largest area is equilateral. How does this result change if we additionally demand that the triangle is right-angled, and of fixed perimeter?

Solution: Maximising the area is equivalent to maximising the area squared, $s(s-a)(s-b)(s-c)$. Hence the Lagrangian is:

$$L(a,b,c,\lambda) = s(s-a)(s-b)(s-c) + \lambda(2s-a-b-c).$$

Extremising, we have:

$$\frac{\partial L}{\partial a} = -sa(s-b)(s-c) - \lambda = 0, \qquad \frac{\partial L}{\partial b} = -sb(s-a)(s-c) - \lambda = 0,$$

$$\frac{\partial L}{\partial c} = -sc(s-a)(s-b) = 0, \qquad \frac{\partial L}{\partial \lambda} = 2s-a-b-c = 0.$$

Equating the first two equations, we have $a(s-b)(s-c) = b(s-a)(s-c)$. Note that $s = c$ would minimise area (it would be zero!), so we can ignore this case. Then $a(s-b) = b(s-a)$. Rearranging, we have:

$$as = bs \qquad \Rightarrow \qquad a = b.$$

Similarly, equating the first and third equations, we have $a = c$, and thus the triangle is indeed equilateral.

If we demand the triangle is right-angled, we could (for example) impose an additional constraint $a^2 + b^2 = c^2$. The new Lagrangian is:

$$L(a,b,c,\lambda,\mu) = s(s-a)(s-b)(s-c) + \lambda(2s-a-b-c) + \mu(a^2+b^2-c^2).$$

Extremising, we now have:

$$\frac{\partial L}{\partial a} = -sa(s-b)(s-c) - \lambda + 2a\mu = 0, \qquad \frac{\partial L}{\partial b} = -sb(s-a)(s-c) - \lambda + 2b\mu = 0,$$

$$\frac{\partial L}{\partial c} = -sc(s-a)(s-b) - 2c\mu = 0, \qquad \frac{\partial L}{\partial \lambda} = 2s-a-b-c = 0, \qquad \frac{\partial L}{\partial \mu} = a^2+b^2-c^2 = 0.$$

15. Find the maximum distance from the origin to the curve $x^2 + y^2 + xy - 4 = 0$.

Solution: We maximise $x^2 + y^2$ subject to the constraint $x^2 + y^2 + xy - 4 = 0$. The Lagrangian is:

$$L(x, y, \lambda) = x^2 + y^2 + \lambda \left(x^2 + y^2 + xy - 4\right).$$

Extremising, we have:

$$\frac{\partial L}{\partial x} = 2x + \lambda(2x + y) = 0, \qquad \frac{\partial L}{\partial y} = 2y + \lambda(2y + x) = 0, \qquad \frac{\partial L}{\partial \lambda} = x^2 + y^2 + xy - 4 = 0.$$

Multiplying the first equation by y, and the second by x, and subtracting, we see that:

$$\lambda y(2x + y) - \lambda x(2y + x) = 0 \qquad \Rightarrow \qquad y^2 = x^2,$$

since if $\lambda = 0$, we would have $x = y = 0$ which does not lie on the curve. Considering both $y = x$ and $y = -x$ in turn then, we have:

- If $y = -x$, substituting into the constraint, we see that $x^2 - 4 = 0$, giving $x = \pm 2$, and hence $y = \mp 2$. This would result in the extremal distance being $\sqrt{8}$.
- If $y = x$, substituting into the constraint, we see that $3x^2 - 4 = 0$, giving $x = \pm\sqrt{4/3}$, and hence $y = \pm\sqrt{4/3}$. This would result in the extremal distance being $\sqrt{8/3}$.

Note that $\sqrt{8} > \sqrt{8/3}$, hence indeed the maximum distance is $\sqrt{8}$.

16.

(a) Extremise the function:

$$f(x_1, ..., x_n) = -\sum_{i=1}^{n} x_i \log(x_i), \qquad \text{subject to} \qquad \sum_{i=1}^{n} x_i = 1.$$

(b) Show that if a further constraint:

$$\sum_{i=1}^{n} x_i y_i = Y$$

is applied, where $y_1, ..., y_n$ and Y are given constants, the extremal point of the same function f is located instead where $x_i = a \exp(-by_i)$ where a and b are constants. Write down two equations that determine the values of a and b. [*You need not solve these equations.*]

Solution:

(a) The Lagrangian is given by:

$$L(x_1, ..., x_n, \lambda) = -\sum_{i=1}^{n} x_i \log(x_i) + \lambda \left(\sum_{i=1}^{n} x_i - 1 \right).$$

Taking the derivative with respect to the kth coordinate, we see that we require:

$$\frac{\partial L}{\partial x_k} = -\log(x_k) - 1 + \lambda = 0$$

for each $k = 1, ..., n$. Thus $x_k = e^{\lambda - 1}$, so that x_k is independent of k. Substituting into the constraint, we see that $ne^{\lambda-1} = 1$, so $e^{\lambda-1} = 1/n$. This gives $x_k = 1/n$ for all k.

(b) With the additional constraint, the Lagrangian is now modified to:

$$L(x_1, ..., x_n, \lambda, \mu) = -\sum_{i=1}^{n} x_i \log(x_i) + \lambda \left(\sum_{i=1}^{n} x_i - 1 \right) + \mu \left(\sum_{i=1}^{n} x_i y_i - Y \right).$$

Taking the derivative with respect to the kth coordinate, we see that we require:

$$\frac{\partial L}{\partial x_k} = -\log(x_k) - 1 + \lambda + \mu y_k = 0$$

for each $k = 1, ..., n$. Therefore $x_k = e^{\lambda-1} e^{\mu y_k}$. Defining $a = e^{\lambda-1}$ and $b = -\mu$, we see that indeed $x_k = ae^{-by_k}$ for some constants a, b, as required.

Substituting into the constraints, we have:

- $\sum_{i=1}^{n} ae^{-by_i} = 1;$
- $\sum_{i=1}^{n} ae^{-by_i} y_i = Y.$

These equations constrain the values of a, b.

17. The function:

$$f(n_0, n_1, n_2, ...) = -\sum_{k=0}^{\infty}[n_k \ln(n_k) - n_k]$$

of an infinite number of positive variables is subject to two constraints:

$$\sum_{k=0}^{\infty} n_k = N \qquad \text{and} \qquad \sum_{k=0}^{\infty} E_0 \left(\frac{1}{2} + k\right) n_k = E,$$

where N, E_0 and E are positive constants. Using the method of Lagrange multipliers, show that the stationary point of $f(n_0, n_1, n_2, ...)$ subject to the above constraints occurs when:

$$n_k = 2N \sinh\left(\frac{\beta E_0}{2}\right) e^{-\beta E_0(1/2+k)}$$

for all k, where β is a Lagrange multiplier. Show further that:

$$E = \frac{NE_0}{2} \coth\left(\frac{\beta E_0}{2}\right).$$

Solution: The Lagrangian is:

$$L(\lambda, \mu, n_0, n_1, n_2, ...) = -\sum_{k=0}^{\infty}[n_k \ln(n_k) - n_k] + \lambda\left(\sum_{k=0}^{\infty} n_k - N\right) + \mu\left(\sum_{k=0}^{\infty} E_0\left(\frac{1}{2} + k\right) n_k - E\right).$$

Taking the derivative with respect to n_i, we see that for each $i = 0, 1, 2, ...$ we require at the stationary point:

$$\frac{\partial L}{\partial n_i} = -\ln(n_i) + \lambda + \mu E_0\left(\frac{1}{2} + k\right) = 0.$$

Rearranging, we see that:

$$n_i = e^{\lambda} e^{\mu E_0(1/2+i)}.$$

Setting $\beta = -\mu$, we indeed see that β is a Lagrange multiplier. It remains to determine λ, which we can do using the first constraint:

$$N = e^{\lambda} \sum_{k=0}^{\infty} e^{-\beta E_0(1/2+k)} = \frac{e^{\lambda} e^{-\beta E_0/2}}{1 - e^{-\beta E_0}}.$$

Rearranging, we have:

$$e^{\lambda} = \frac{(1 - e^{-\beta E_0})N}{e^{-\beta E_0/2}} = \left(e^{\beta E_0/2} - e^{-\beta E_0/2}\right) N = 2N \sinh\left(\frac{\beta E_0}{2}\right).$$

Hence, we indeed have the required expression for n_k:

$$n_k = 2N \sinh\left(\frac{\beta E_0}{2}\right) e^{-\beta E_0(1/2+k)}.$$

Substituting into the second constraint, we have:

$$E = 2NE_0 \sinh\left(\frac{\beta E_0}{2}\right) \sum_{k=0}^{\infty}\left(\frac{1}{2} + k\right) e^{-\beta E_0(1/2+k)}.$$

We can write this sum as:

$$\sum_{k=0}^{\infty}\left(\frac{1}{2}+k\right)e^{-\beta E_0(1/2+k)} = -\frac{1}{E_0}\frac{d}{d\beta}\sum_{k=0}^{\infty}e^{-\beta E_0(1/2+k)} = -\frac{1}{E_0}\frac{d}{d\beta}\frac{e^{-\beta E_0/2}}{1-e^{-\beta E_0}}.$$

Observe that:

$$\frac{e^{-\beta E_0/2}}{1-e^{-\beta E_0}} = \frac{1}{e^{\beta E_0/2}-e^{-\beta E_0/2}} = \frac{1}{2\sinh(\beta E_0/2)} = \operatorname{cosech}\left(\frac{\beta E_0}{2}\right).$$

Taking the derivative, we have:

$$-\frac{1}{2E_0}\frac{d}{d\beta}\operatorname{cosech}\left(\frac{\beta E_0}{2}\right) = \frac{1}{4E_0}\coth\left(\frac{\beta E_0}{2}\right)\operatorname{cosech}\left(\frac{\beta E_0}{2}\right).$$

Substituting back into the constraint for E, we have:

$$E = \frac{NE_0}{2}\coth\left(\frac{\beta E_0}{2}\right),$$

as required.